PART I: PROBLEM STATEMENT AND RISK-COST ANALYSIS

Question 1.1: Clinical problem statement and data space (1.0 point)

1. **Target Patient Population:**
    - Adult and pediatric patients visiting medical facilities (such as emergency rooms or outpatient clinics) who present with respiratory symptoms (e.g., cough, fever, shortness of breath) and require diagnostic chest radiography.
2. **Input Feature Space (X) and Output Label (Y):**
    - **Input Feature Space (X):** Preprocessed chest X-ray images, represented as pixel matrices or deep feature tensors.
    - **Output Label (Y):** A binary classification label where $Y \in \{0,1\}$ (0 denotes Normal/No Finding, and 1 denotes Pneumonia).
3. **Machine Learning Problem Classification:**
    - **Supervised Learning:** Because the model is trained based on pre-assigned labels (Normal and Pneumonia) in the dataset.
    - **Binary Classification:** Because the problem only divides the results into two classes (e.g., Normal or Pneumonia)

Question 1.2: Asymmetric risk-cost analysis and Metric Selection (1.0 point)

1. **Asymmetric Cost Analysis (FP vs. FN):**
    - **False Positive (FP):** The model predicts pneumonia in a healthy patient. Causes psychological distress, incurs additional costs for unnecessary confirmatory testing, and overburdens clinical staff.
    - **False Negative (FN):** The model predicts a healthy status for a patient actually suffering from pneumonia. Extremely dangerous medically, leads to delayed treatment, rapid disease progression, severe respiratory failure, or potentially fatal outcomes.
    - False Negative (FN) is significantly more dangerous because missing a life-threatening infection directly jeopardizes patient survival and health.
2. **Target Metric Selection:**
    - **Primary Metric: Recall (Sensitivity)** or **PR-AUC (Precision-Recall Area Under Curve)**, alongside Specificity constraints.
    - **Clinical Justification:** A False Negative poses a lethal risk, the model must prioritize catching as many positive cases as possible, making high Sensitivity essential. Because medical datasets suffer from class imbalance, PR-AUC and Sensitivity provide a more robust performance evaluation, preventing the model from hiding behind majority-class bias.

Question 2.1: Design and validation of the literature review matrix (3.0 points)

1. **Mandatory Requirements for Journal Ranking, Publication Year, Conference Papers, and Preprints:**
    - **Journal Ranking & Year:** All 10 selected papers must be published between 2020 and 2026 and indexed in high-ranking peer-reviewed journals classified as SJR Q1 or Q2
    - **Conference Papers:** Top-tier peer-reviewed artificial intelligence and medical imaging conferences are fully accepted.

- **Preprints (arXiv):** arXiv preprints are strictly prohibited from being used as primary baseline references unless they have been formally accepted and published in a peer-reviewed Q1/Q2 journal or top conference proceedings.

2. **The 8 Mandatory Information Fields in the Literature Comparison Matrix:**

   Paper Title & Authors, Publication Venue & Year, Clinical Application Domain & Objective, Dataset Characteristics & Size, Input Feature Space & Preprocessing Pipeline, Model Architecture / Core Algorithm, Data Split Protocol, Reported Results & Metrics

3. **Traceability Requirement:**

   - Every quantitative metric, architectural choice, or dataset specification recorded in the matrix must include explicit traceability metadata pointing directly to the original paper, specifically indicating the Exact Page Number, Table Number, or Figure Number where the data is published

Question 2.2: Research gap analysis and baseline determination (1.0 point)

1. **Research Gaps Analysis via Cross-Referencing:**

   - **Cross-Referencing Method:** Researchers cross-reference the "Data Split Protocol" column against the "Reported Results" column to detect systemic evaluation flaws or vulnerabilities.

   - **Gap 1 (Patient-Level Leakage Flaw):** Cross-referencing reveals that studies using random image-level splits report inflated accuracies ($\sim$ 99%). When evaluated on independent multi-center patient cohorts, their generalization collapses, highlighting a critical lack of robust cross-institution validation.

   - **Gap 2 (Shortcut Learning Vulnerability):** Cross-referencing performance metrics with qualitative Grad-CAM or occlusion analysis across studies shows high classification accuracy tied to non-clinical features rather than pathological lung infiltrates.

2. **Baseline Models Determination:**

   - **ResNet-50**: Standard residual deep learning architecture widely adopted as a benchmark for medical image classification.

   - **DenseNet-121** (Historically proven and heavily cited backbone specifically optimized for chest X-ray multi-label and binary pathology classification tasks).

Question 2.3: Citation management and anti-hallucination rules (1.0 point)

1. **Single Source of Truth (SSoT) Directory and Naming Convention:**

   - **Directory Structure:** All PDF files of cited papers must be stored in a centralized, version-controlled local directory designated as the Single Source of Truth (SSoT) repository (e.g., ./literature_ssot_repository/).

   - **File Naming Convention:** Files must follow a strict standardized format: [AuthorLastName]_[Year]_[FirstThreeWordsOfTitle].pdf (e.g., Johnson_2023_DeepLearningPneumonia.pdf). Sub-folders can categorize papers by core themes (e.g., /shortcut_learning/, /chest_xray_ai/).

2. **Penalty Regulations for Violations:**

   - **Missing PDF File:** If a cited paper in the matrix or report cannot be verified with a corresponding stored PDF file inside the SSoT repository, it is treated as an unverified citation, resulting in an automatic **zero score** for the literature review section.

- **Data Mismatch:** If quantitative data or metadata recorded in the matrix fails to match the original paper upon auditing, or if information is fabricated/hallucinated, severe academic integrity penalties apply, including project disqualification.

PART III: EXPLORATORY DATA ANALYSIS - EDA (3.0 POINTS)

Question 3.1: Specification of Biomedical Data Attributes and Epidemiological Bias (1.5 points)

1. **Core Descriptive Parameters of a Medical Dataset:**
    - **Origin & License:** The project utilizes chest X-ray imaging data compiled from public repositories, specifically the Guangzhou Chest X-Ray dataset and the Kaggle NIH Chest X-rays dataset.
    - **License / Access Conditions:** Open-access research datasets made available via Kaggle for academic benchmarking and computer vision experimentation.
    - **Number of Patients & Samples:** 5,683 images in Chest X-Ray Images (Pneumonia), >112,000 images from 30,805 patients in NIH Chest X-rays.
    - **Scale & Dimensions:** Single-channel grayscale images ($1 \times H \times W$), rescaled and resized to standard input dimensions (e.g., 224×224 pixels for EfficientNet-B0 training
    - **Resolution & Sampling:** Varying native digital radiography resolutions, standardized during the preprocessing pipeline.
2. **Specific Demographic Profiles of the Datasets**

- **Guangzhou Chest X-Ray Dataset (Pneumonia Detection):**
    - **Age Distribution:** The dataset exclusively comprises pediatric patients aged 1 to 5 years old selected from retrospective cohorts at the Guangzhou Women and Children's Medical Center.
    - **Gender/Class Confounding:** The subjects are entirely young children, meaning adult data is completely absent from this training source.
- **NIH Chest X-Ray Dataset:**
    - **Age & Gender Distribution:** Contains over 112,000 frontal-view images from 30,805 unique adult and pediatric patients across a wide spectrum of ages (ranging from newborns to elderly individuals aged over 90) and varying gender distributions. Exploratory statistics typically show higher representation in middle-aged adult groups (e.g., ages 40–60).

Direct Impact on Model Training and Evaluation Biases

- **Severe Age-Group Mismatch and Domain Shift:**
    - Because the Guangzhou dataset is strictly limited to children (1–5 years old), an EfficientNet model trained on it learns anatomical features unique to pediatric thoracic structures (e.g., higher thymus visibility, different rib cage-to-lung volume ratios, and distinct cardiac-to-thoracic width proportions).
    - If this model is evaluated or deployed on adult data from the NIH dataset, it will suffer from a severe domain shift and generalization failure. The model will misinterpret normal adult anatomical variations as pathological abnormalities because it has never encountered adult baseline imagery.
- **Anatomical Shortcut Learning Induced by Age/Size:**
    - In pediatric chest X-rays, the physical dimensions of the rib cage, bone density, and organ positioning are scaled differently compared to adults.

- If the age distribution between the "Normal" and "Pneumonia" subsets within either dataset is imbalanced (e.g., pneumonia cases clustered heavily within a specific narrow age bracket), the deep learning model exploits age-correlated skeletal geometry or size metrics (as observed in your Grad-CAM highlighting ribs) rather than true infection opacities. This leads to artificially inflated validation metrics that collapse in real-world clinical deployment.

3. **Statistical Features for Imaging Data (Brightness Histogram):**

   In the context of the chest X-ray classification project utilizing the Guangzhou pediatric dataset and the NIH adult dataset, the specific imaging features, statistical metrics, and corresponding chart types are defined as follows:

- **Specific Statistical Features of the Images:**
  - **Pixel Intensity & Grayscale Amplitude Statistics:** X-ray images are represented as 2D matrices where pixel values correspond to tissue radiodensity. The key statistical features extracted and monitored include the mean intensity, standard deviation, and dynamic range of grayscale pixel values.
  - **Brightness/Contrast Histograms:** The distribution of pixel intensities across regions of interest (lung parenchyma versus bone structures like ribs and clavicles).
  - **Spatial Dimensions & Aspect Ratios:** Pixel height and width distributions across different imaging devices (e.g., varying scanning resolutions between Guangzhou and NIH cohorts).
- **Corresponding Chart Types and Project Application:**
  - **Grayscale Intensity and Contrast Histograms:**
    - Application: Used to analyze the pixel value distribution before and after applying preprocessing techniques like CLAHE (Contrast Limited Adaptive Histogram Equalization) in your pipeline.
    - Specific Impact: Chest X-rays suffer from inconsistent exposure settings and X-ray tube current variations, resulting in shifted brightness histograms. Plotting histograms can help verify whether CLAHE effectively normalizes local contrast across heterogeneous scans, preventing the model from utilizing background exposure levels or scanner-specific artifacts as classification shortcuts.
  - **Lung Region Bounding Box and Aspect Ratio Distribution Charts:**
    - Application: Visualizing the spatial distribution of lung bounding boxes or mask crop coordinates (get_lung_mask_and_crop).
    - Specific Impact: Ensures that the preprocessing step successfully eliminates peripheral noise, text labels, and external body contours (which often trap Grad-CAM attention onto ribs or skin edges), forcing the statistical feature extraction exclusively onto the thoracic cavity and lung fields.

Question 3.2: Analysis of Medical Data Challenges (1.5 points)

1. Class Imbalance: Sample-level vs. Patient-level Imbalance in the Datasets

- Sample-level Imbalance:
  - The ratio calculated by comparing the total number of individual image files (scans) belonging to the positive class versus the negative class.
  - In the Guangzhou Chest X-Ray dataset, the sample-level distribution is heavily skewed, containing significantly more Pneumonia scans than Normal scans (or vice versa depending

on the specific training/validation split folders). Each image file is treated independently, ignoring how many scans originate from the same subject.

- Patient-level Imbalance:
    - The ratio calculated by grouping scans by unique patient IDs (metadata identifiers) to see how many individual patients have positive diagnoses versus negative diagnoses.
    - This is critically important in the NIH Chest X-Ray dataset (Data_Entry_2017.csv), where a single patient can contribute multiple sequential or varying-view X-ray images over time. If patient-level splitting is not enforced, multiple images from the same patient can leak across both the training and validation sets, causing severe data leakage, artificially inflated model accuracy, and masking true patient-level demographic or disease prevalence imbalances.

2. Specific Data Quality, Noise, and Duplicate Issues in Chest X-Ray Datasets

- Anatomical and Positional Noise:
    - Chest X-rays in both the Guangzhou pediatric cohort and the NIH adult dataset frequently suffer from patient motion blur during acquisition, incorrect thoracic positioning (rotated spines), and variable inspiration levels (incomplete expansion of lung fields), which alter the geometric appearance of the heart and rib cage.
- Imaging and Equipment Artifacts (High-Contrast Noise):
    - Presence of external metallic objects and baked-in textual radiology markers (left/right orientation tags or hospital ID stamps) directly on the image matrix. These act as high-contrast noise or shortcut features that deep learning models mistakenly lock onto during training.
- Label Noise and Diagnostic Ambiguity in NIH Multi-Label Data:
    - The NIH dataset utilizes text-mining of radiology reports to generate labels, which introduces natural label noise (FP/FN). Furthermore, conditions like early-stage pneumonia can overlap visually with atelectasis, pleural effusion, or normal pulmonary vasculature, leading to ambiguous ground-truth labels.
- Duplicate and Near-Duplicate Samples:
    - The NIH dataset contains multiple near-duplicate instances or repeated scans of the same patient taken moments apart with identical presentation, risking severe data contamination if train/test splits are performed randomly instead of strictly isolating patient IDs.

3. Penalty Regulations for Missing EDA Source Code or Non-Runnable End-to-End Code

- Automatic Score Deduction / Zero Grade Policy:
    - Reports lacking an Exploratory Data Analysis (EDA) source code link or containing source code that fails to execute successfully from end-to-end will face severe penalties, up to a complete forfeiture of points for the programming and implementation sections.

PART IV: ADVANCED — PREPROCESSING AND ZERO DATA LEAKAGE (BONUS +2.0 POINTS)

Question 4.1: Domain-specific biomedical data preprocessing (+1.0 point)

1. **Hounsfield Unit (HU) Windowing:**
    - Applied primarily to CT scans to clip and scale pixel intensity values to a specific clinical window of interest (e.g., lung window or soft tissue window). This filters out irrelevant

anatomical structures (such as bones or surrounding air) and enhances the contrast of target pathologies.

2. **CLAHE (Contrast Limited Adaptive Histogram Equalization):**
    - Applied to X-ray and fundus images to enhance local contrast, especially in low-intensity regions, while preventing over-amplification of noise by limiting the contrast amplification factor within small image tiles.
3. **Intensity Normalization:**
    - Scaling pixel intensity values to ensure numerical stability and consistent feature distributions across different input images during deep neural network training.

Question 4.2: Absolute data leakage prevention discipline — Zero Data Leakage (+1.0 point)

1. **Independent-Unit Splitting:**
    - In medical datasets, a single patient often contributes multiple scans or correlated frames. Random sample-level splitting causes highly similar scans from the same patient to leak into both the training and validation/test sets. The model memorizes patient-specific anatomical quirks rather than learning generalized pathology features, resulting in severely inflated and unreliable performance metrics.
    - **The principle of Patient-Level Splitting & StratifiedGroupKFold:** Data splitting must be strictly grouped by a unique identifier (patient_id, subject_id, or lesion_id). All images belonging to the same patient must reside entirely within a single split (Train, Validation, or Test). StratifiedGroupKFold is utilized to maintain class distribution proportions across folds while ensuring absolute patient-level independence.
2. **Fit-on-Train-Only Rule:**
    - **Rule application:** All data-dependent operations—such as scalers (StandardScaler), imputers (KNNImputer), encoders, feature selection algorithms, and oversampling techniques (e.g., SMOTE)—must be **fitted on the training set**.
    - **Before vs. After splitting:**
        - *Before splitting:* Dataset cleaning, deduplication, and patient-level grouping/partitioning.
        - *After splitting (Fit-on-Train-Only):* Parameters learned from the training fold are subsequently *applied* (transformed) to the validation and test sets without re-fitting, preventing any leakage of statistical information from evaluation sets into the training process.